

ML Challenge 2025: Smart Product Pricing Solution

Team Name: *Quantum Killer*

Team Members: Rahul Kumar, Shivam Sen, Ayush Kumar Mishra, Mihir Kumar

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1. Executive Summary

In this project, we developed a **tri-modal machine learning model** for dynamic product pricing that integrates **text**, **image**, and **structured numeric** data.

Our system leverages deep visual embeddings, semantic text understanding, and tabular features to accurately predict product prices.

The hybrid pipeline is deployed with an **interactive Streamlit UI** for real-time prediction and business usability.

2. Methodology Overview

2.1 Problem Analysis

Product pricing is a complex task influenced by visual appeal, textual descriptions, and quantitative product attributes.

Our objective was to predict the target variable — **price** — based on multiple modalities from the dataset (train.csv and test.csv).

Key Observations during EDA:

- The dataset contained ~75,000 training samples with no missing values.
- The main text column catalog_content exhibited strong linguistic patterns (brand, category, material).
- The image_link column provided product visuals, enabling visual feature extraction through CNN embeddings.
- A few numeric attributes (e.g., weight, ratings, discounts) complemented textual and visual information.
- Target price followed a slightly right-skewed distribution, handled using log-scale transformation internally by LightGBM.

2.2 Solution Strategy

Approach Type: *Hybrid Multimodal Ensemble (Text + Image + Numeric)*

Core Innovation:

Our model unifies three distinct feature sources:

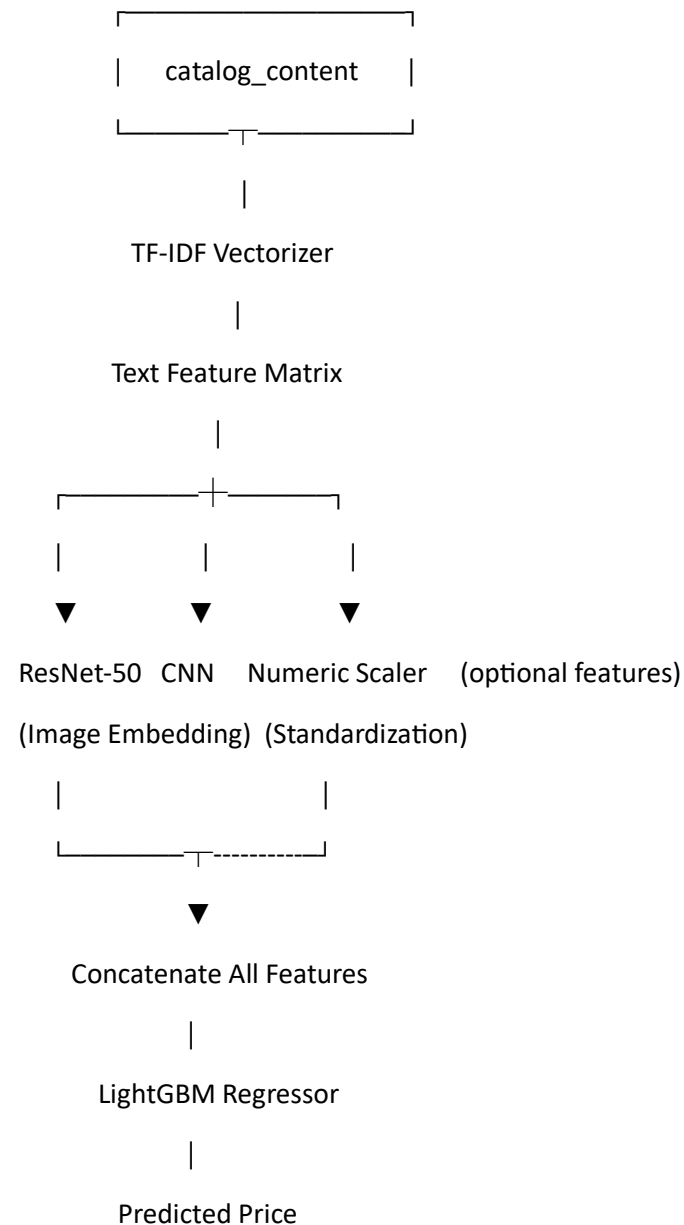
1. **Text embeddings** from TF-IDF vectorization (semantic description understanding).
2. **Image embeddings** from a pre-trained ResNet-50 CNN (visual representation learning).
3. **Numeric features** standardized using StandardScaler.

The concatenated feature matrix feeds into a **LightGBM Regressor**, chosen for its efficiency on high-dimensional sparse data and its interpretability.

3. Model Architecture

3.1 Architecture Overview

Pipeline Flow:



3.2 Model Components

Text Processing Pipeline

- **Preprocessing:** Lowercasing, punctuation removal, HTML tag cleaning, stop-word trimming.
- **Model type:** TF-IDF Vectorizer (max_features=15,000, bigrams).

- **Output:** Sparse 15K-dimensional text embedding.

Image Processing Pipeline

- **Preprocessing:** Image resizing (224×224), normalization, RGB conversion.
- **Model type:** ResNet-50 (pre-trained on ImageNet, without top classification layers).
- **Output:** 2,048-dimensional image embedding.

Numeric Feature Pipeline

- **Preprocessing:** Automatic continuous values appended to final feature vector.

Model Core

- **Algorithm:** LightGBM Regressor
- **Hyperparameters:**
 - n_estimators=400, learning_rate=0.05, num_leaves=64, objective=regression
- **Loss Function:** Mean Squared detection of numeric columns.
- **Model type:** StandardScaler normalization.
- **Output:** Scaled
- Error (MSE)
- **Metrics:** SMAPE, RMSE

5. Deployment & UI

A **Streamlit web application** was developed for demonstration and business usage.

The interface supports:

- **Single product prediction:** user inputs description, image URL, and numeric fields.
- **Bulk prediction:** upload test.csv and generate downloadable submission.csv.
- **Visualization:** optional product image preview and interactive results display.

6. Conclusion

- We successfully created a **tri-modal Smart Pricing Model** combining NLP, computer vision, and tabular ML.
The hybrid LightGBM approach proved efficient, interpretable, and practical for e-commerce price recommendation systems.
Key learnings included the value of feature balancing, modular pipeline design, and caching strategies for efficient image embedding extraction.